



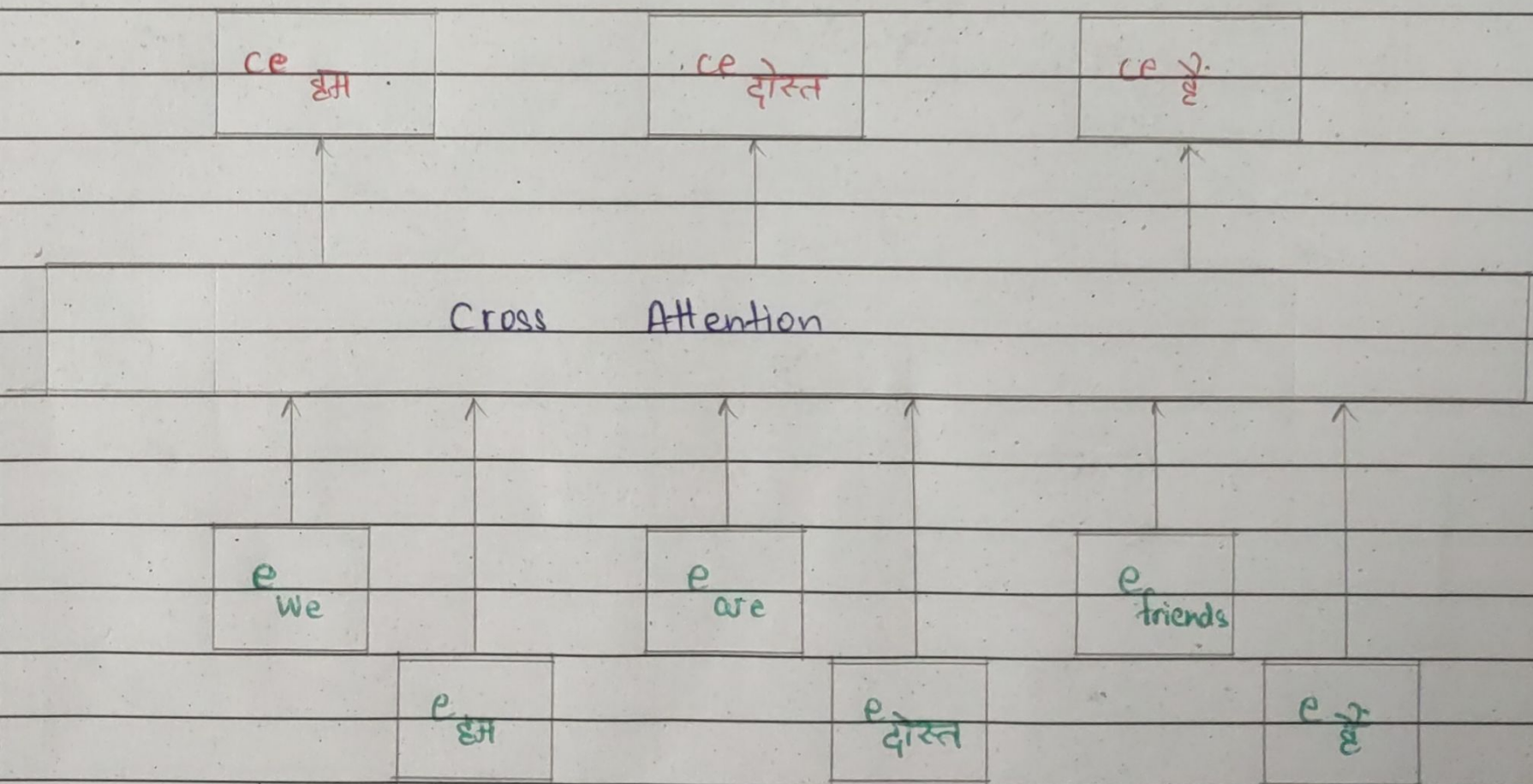
Consider the task of machine translation where the input sequence is "we are friends" & output sequence is "हम दोस्त हैं".

Input

Cross attention tries to derive a relationship between input & output sequences thus it takes embeddings of both sequences as input.

Output

Cross attention outputs contextual embeddings of output sequences, which derive the relationship between input & output sequences. The no. of contextual embeddings generated from Cross Attention is exactly same as no. of words in the output sequence.

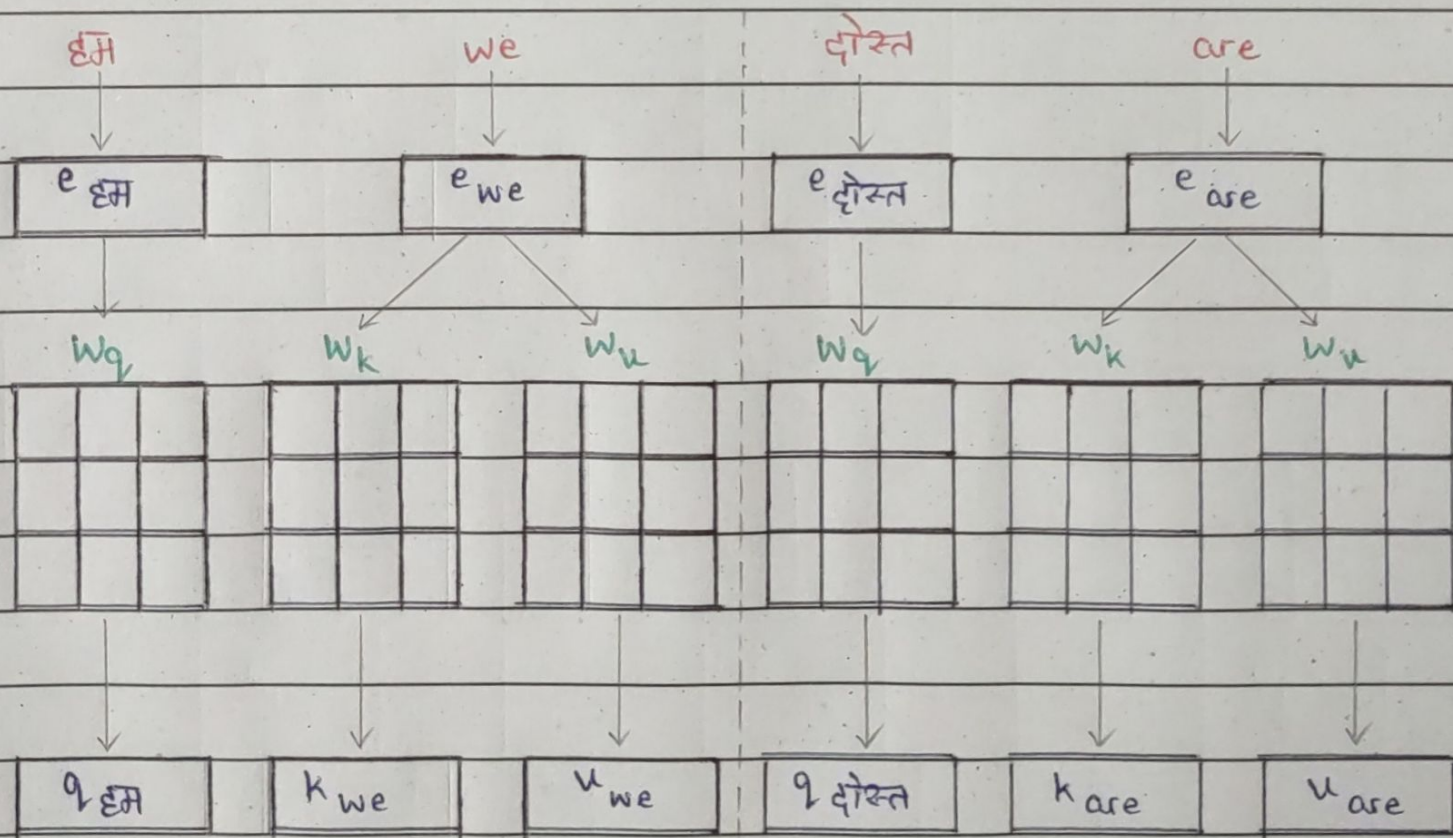


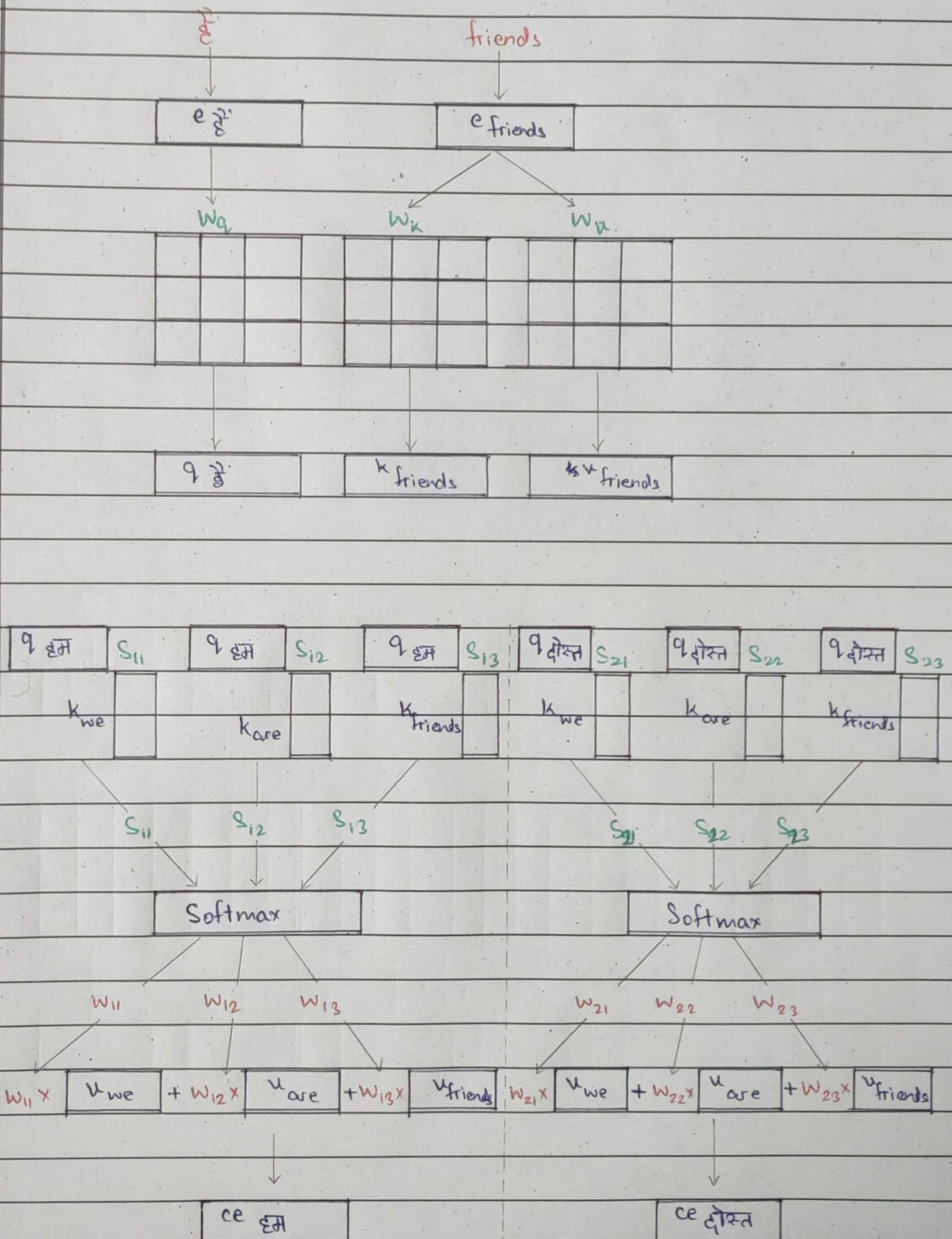


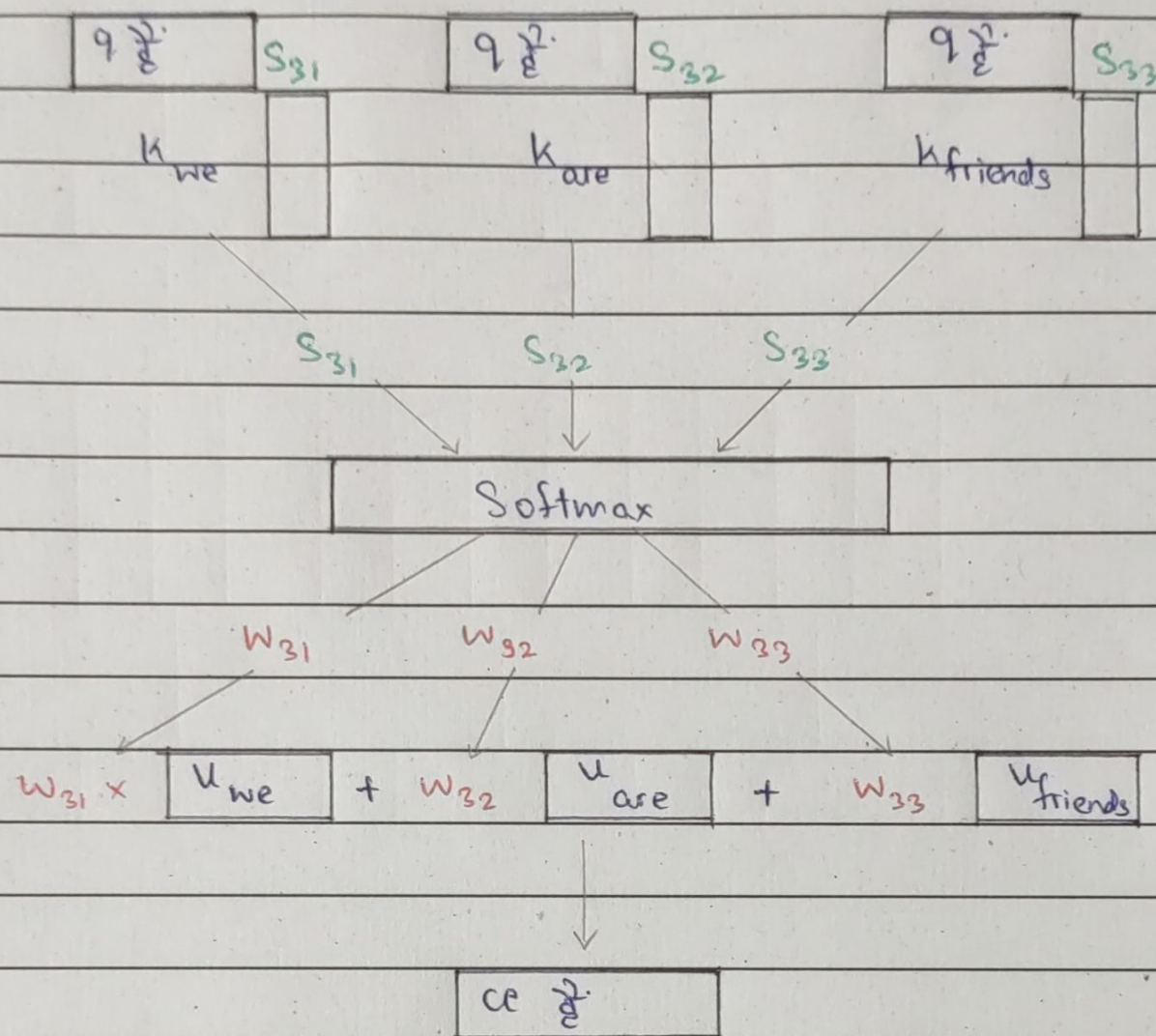
Processing

In cross attention both embeddings of input & output sequences are passed. The embeddings of output sequences are multiplied with Query weight matrix to generate query vectors whereas the embeddings of output input sequences are multiplied with Key & Value matrices to generate key & value vectors.

The query vectors of output sequences are multiplied with key vectors of input sequences the resultants of which are passed through a softmax layer which generates normalized weights which are then used as a weighted to calculate weighted sum using the value vectors which is the resultant contextual embeddings of the output sequence.







The embeddings of input sequences that are passed into the cross attention are the contextual embeddings of the input sequences generated from the Encoder architecture.

The embeddings of output sequences are masked contextual embeddings generated by the Masked Self Attention inside the decoder in the previous step.